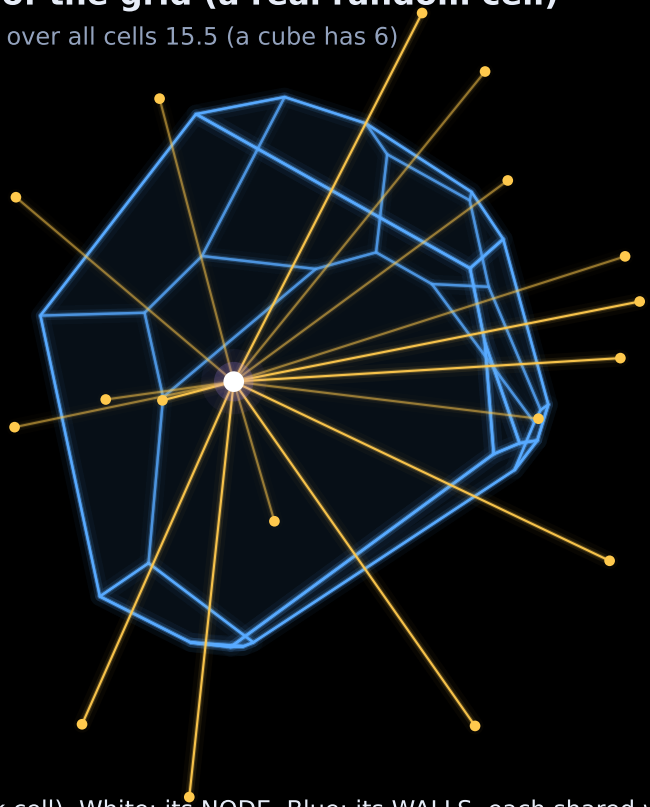


Figure 10. The grid's building block: a random cell, its links, and the size it must have

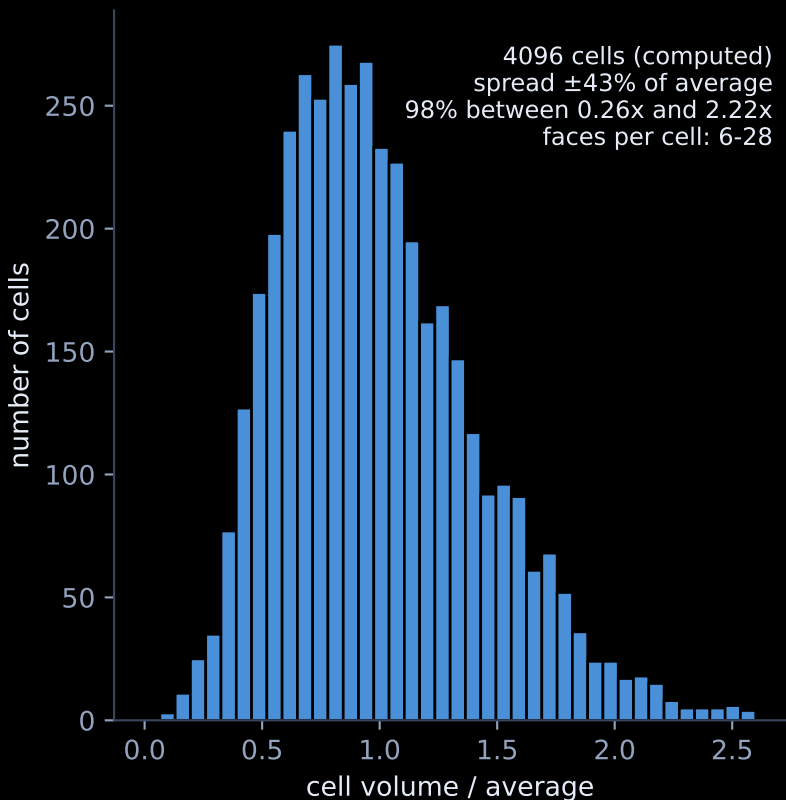
(a) One cell of the grid (a real random cell)

17 faces; average over all cells 15.5 (a cube has 6)

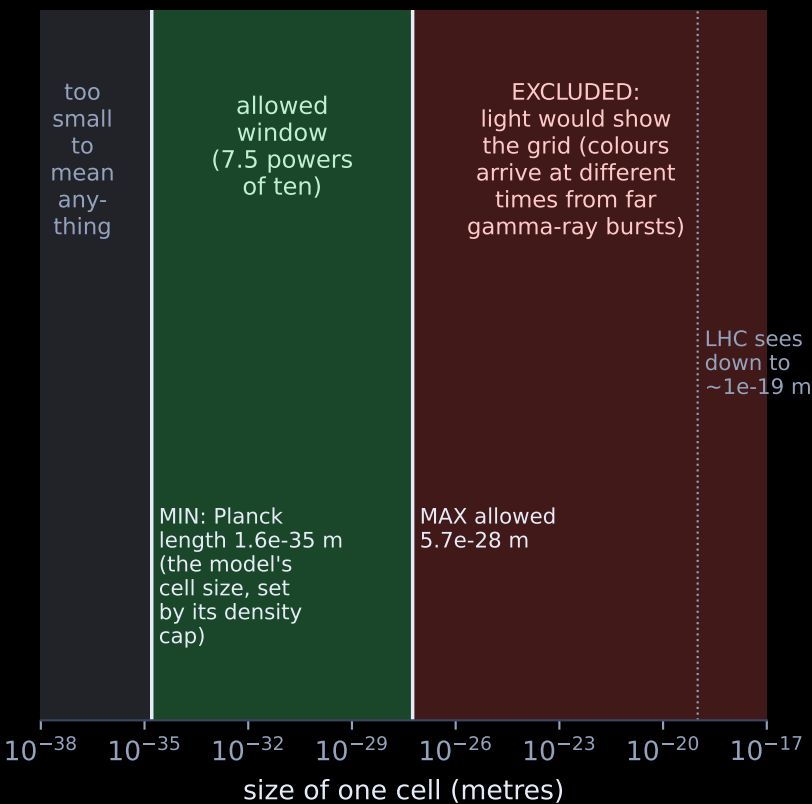


GRID CELL (Planck cell). White: its NODE. Blue: its WALLS, each shared with one neighbouring cell. Gold: its TENSION LINKS, one through each wall to the neighbouring node; a link's strength is set by its wall's area.

(b) Cells come in a spread of sizes



(c) How big can a cell be?



No cell needs a set size or shape: only the AVERAGE size matters, and randomness is what keeps every direction equal.

Equations: tension link weight $w_{ij} = A_{ij}/d_{ij}$ (wall area / link length); field on the grid $\sum_j w_{ij}(\phi_j - \phi_i) = 4\pi G m_i$

MAX cell size $\ell_{max} = \sqrt{12} \hbar c/E_{QG,2} = 5.7 \times 10^{-28} \text{ m}$ ($E_{QG,2} > 1.2 \times 10^{12} \text{ GeV}$, GRB 221009A); MIN $\ell_p = \sqrt{\hbar G/c^3} = 1.6 \times 10^{-35} \text{ m}$